Q1 How far can the concepts studied so far be generalized?

Q2 What are the salient properties of $\mathbb{R}^n$ that enabled the development of concepts till now?

Axiomatic approach

Number systems

$$\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C}$$

? ?

$\sqrt{2}$ $\sqrt{3}$

$\sqrt{3} = \sqrt{\frac{3}{2}}\,\sqrt{2}$

$\sqrt{3} = (\text{rational})\,\sqrt{2}$] $\leftarrow$

Field (Number field)

A field is a set F with two binary operations

$+$ and $\cdot$

1) Commutativity 2 Associativity:

$a, b \in F$

$a+b = b+a$

$a \cdot b = b \cdot a$

$(a+b)+c = a+(b+c)$

$a\cdot(b\cdot c) = (a\cdot b)\cdot c$

2) Additive & multiplicative identities:

0: $a+0 = a$ for all $a \in F$.

1: $a \cdot 1 = a$ for all $a \in F$

3) Additive & multiplicative inverses:

for every $a \in F$, there is a $-a \in F$ such that $a + (-a) = 0$.

for every $a \in F \setminus \{0\}$, there is a $a^{-1} \in F$ such that $a a^{-1} = 1$.

4) Closure to addition & multiplication:

$a, b \in F, \Rightarrow a+b \in F$, $a \cdot b \in F$.

5) $a(b+c) = ab + ac$ for any $a, b, c \in F$.

Prove that $0 \cdot 1 = 0$ using properties 1) 2) 3a) 4) & 5).

$$\begin{cases} 0 \cdot 1 = \underbrace{(0+0) \cdot 1} = 0 \cdot 1 + 0 \cdot 1 \\ \cancel{0 \cdot 1} + \cancel{(-0 \cdot 1)} = 0 \cdot 1 + \cancel{0 \cdot 1} + \cancel{(-0 \cdot 1)} \\ 0 \qquad\qquad\qquad\qquad 0 \end{cases}$$

$0 \cdot a$
$= (0+0) \cdot a$
$= 0 \cdot a + 0 \cdot a$

$\mathbb{N}$; $\mathbb{N} \cup \{0\}$, $\mathbb{Z}$ not fields.

$\mathbb{Q}$ is a field: (set of rational numbers).

$\mathbb{R}$ is a field.

$\mathbb{C}$ is also a field.

Boolean field: $\{0, 1\}$

$0 + 0 = 0$

(XOR)

+	0	1

$\odot$	0	1

0	0	1
1	1	0

0	0	0
1	0	1

If $0 \equiv$ FALSE & $1 \equiv$ TRUE, the addition in boolean field is XOR & multiplication is AND.

If the addition were OR, then 1 fails to have an additive inverse. Hence $\{0, 1\}$ fails to be a field with OR.

$1+1=1$

$1+0=1$

Modular fields

Can interpret boolean field above with

$0 \equiv$ even $\qquad 1 \equiv$ odd.

Addition & multiplication tables are filled assuming field elements are reminders modulo p.

p=2

p=3

$\{0, 1, 2\}$

$(3n+2)(3m+2)$
$= 3\square + 1$

$(-1) = 2$

$(-2) = 1$

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

$1^{-1} = 1$

$2^{-1} = 2$

$p = 4$ $\quad \{0,1,2,3\}$

Exer:

Make the addition table

$3+1 = 0$

$3+2 = 1$

$3+0 = 3$

$3+3 = 2$

$\cdot$	0	1	2	3
0				
1				
2	0	2	0	2
3				

Not a field

2 does not have multiplicative inverse

Example:

$S = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$

Is S a field?

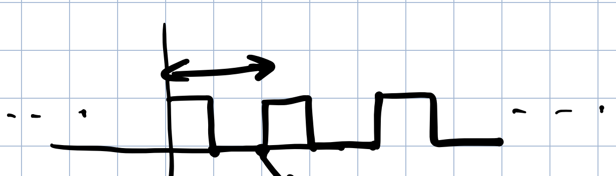
$$\frac{1}{a+b\sqrt{2}} = \frac{a-b\sqrt{2}}{a^2-2b^2}$$

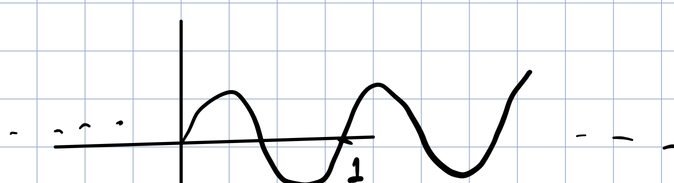
$a \cdot b = 0$

$a^{-1} a \cdot b = \underbrace{a^{-1} \cdot 0}_{0}$

Consider the space of functions $f : \mathbb{R} \to \mathbb{R}$ that are periodic with period 1.

$f(t+1) = f(t)$ for every t.





1

we will see that this space of functions is a vector space

Vector space over a number field K → typically $\mathbb{R}$ or $\mathbb{C}$

Is a set V with two operations $\alpha\,\underline{v}$

1) Addition of two vectors $+$

2) Multiplication of "scalar" with vector.

Such that $\mathrm{Sum}(\underline{v}_1, \underline{v}_2)$

(i) Commutative $\underline{v}_1 + \underline{v}_2 = \underline{v}_2 + \underline{v}_1$ $\quad f_1+f_2 = f_2+f_1$ $\quad \underline{v}_3$

(ii) Associative $(\underline{v}_1 + \underline{v}_2) + \underline{v}_3 = \underline{v}_1 + (\underline{v}_2 + \underline{v}_3)$

$= \underline{v}_1 + \underline{v}_2 + \underline{v}_3$

(iii) Additive identity & inverse:

$\underline{v}_1 + \underline{0} = \underline{v}_1$ $\quad \underline{0}$: $f(t) = 0$ for all $t \in \mathbb{R}$.

$\underline{v}_1 + (-\underline{v}_1) = 0$.

(iv) closure to addition, additive inverse & scaling.

(v) $\alpha(\beta\,\underline{v}) = (\alpha\beta)\,\underline{v}$ for any $\alpha, \beta \in K$, $\underline{v} \in V$.

(vi) $(\alpha+\beta)\,\underline{v} = \alpha\underline{v} + \beta\underline{v}$

(vii) $1\,\underline{v} = \underline{v}$ for all $\underline{v}$.

Prove that $0 \cdot \underline{v} = \underline{0}$ for any $\underline{v} \in V$ using the properties above.

Exercise.

Example: the space of periodic functions $f: \mathbb{R} \to \mathbb{R}$ with period 1 discussed above form a vector space over $\mathbb{R}$.

$f(t) = g(t)$
for all $t \in \mathbb{R}$
then $f = g$.

If f & g are periodic with period 1
then $\underline{f(t) + g(t)}$ is also periodic with period 1.

$h(t) = f(t) + g(t)$

$\underbrace{h(t+1)} = f(t+1) + g(t+1)$

$= f(t) + g(t)$

$= h(t)$.

Likewise, the space of all functions $f: \mathbb{R} \to \mathbb{R}$ is a vector space over $\mathbb{R}$.

Result: The functions $\{\sin 2\pi k x\}_{k=1}^{\infty}$, $\{\cos 2\pi k x\}_{k=0}^{\infty}$ are a <u>BASIS</u> of the space of all periodic functions of period 1.

(Spanning)
$$\underbrace{f(t)}_{\text{period } 1} = \left. \sum_{k=0}^{\infty} a_k \cos 2\pi k x + \sum_{k=1}^{\infty} b_k \sin 2\pi k x \right\}$$

Fourier series.

$a_1 \sin 2\pi x + a_2 \sin 4\pi x = 0$ for all $x \in \mathbb{R}$.

$a_1 \sin 2\pi x + a_2 \sin 4\pi x + a_3 \sin 6\pi x = 0$.

Exercise.

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Inner product:

between two functions as

$$\int_{-\infty}^{\infty} f(t)\, g(t)\, dt$$

(for arbitrary

functions)

(i) $$\int_{-\infty}^{\infty} \big(f_1(t)+f_2(t)\big)\, g(t)\, dt = \int_{-\infty}^{\infty} f_1(t)\, g(t)\, dt + \int_{-\infty}^{\infty} f_2(t)\, g(t)\, dt$$

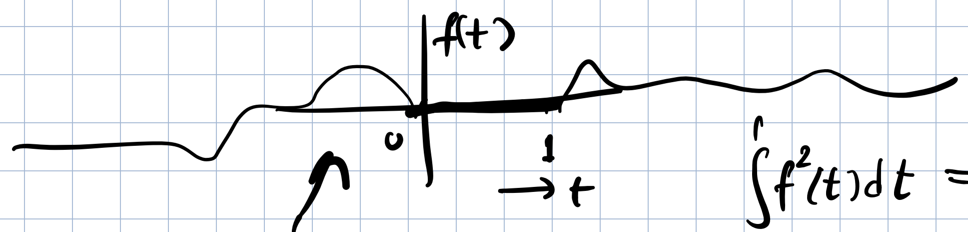
(ii) check other properties of inner product — (Exercise)

The vectors in the basis $\{\sin 2\pi k x\}_{k=1}^{\infty}$ $\{\cos 2\pi k x\}_{k=0}^{\infty}$ are orthogonal.

$$\int_0^1 \sin 2\pi x \cos 2\pi x \, dx =$$

(For periodic functions, of period 1) $$\int_0^1 f(t)\, g(t)\, dt \quad \text{— (1)}$$

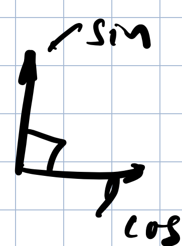
(iv) $\langle f, f\rangle \geq 0$ and $\langle f, f\rangle = 0$ only when $f=0$

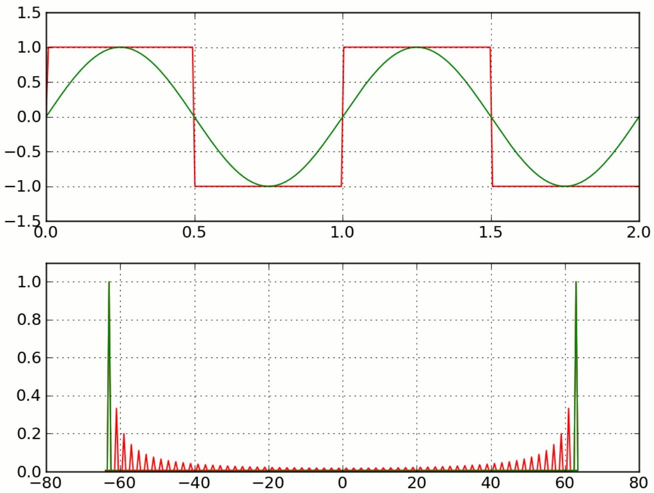


$$\int_0^1 f^2(t)\, dt = 0$$

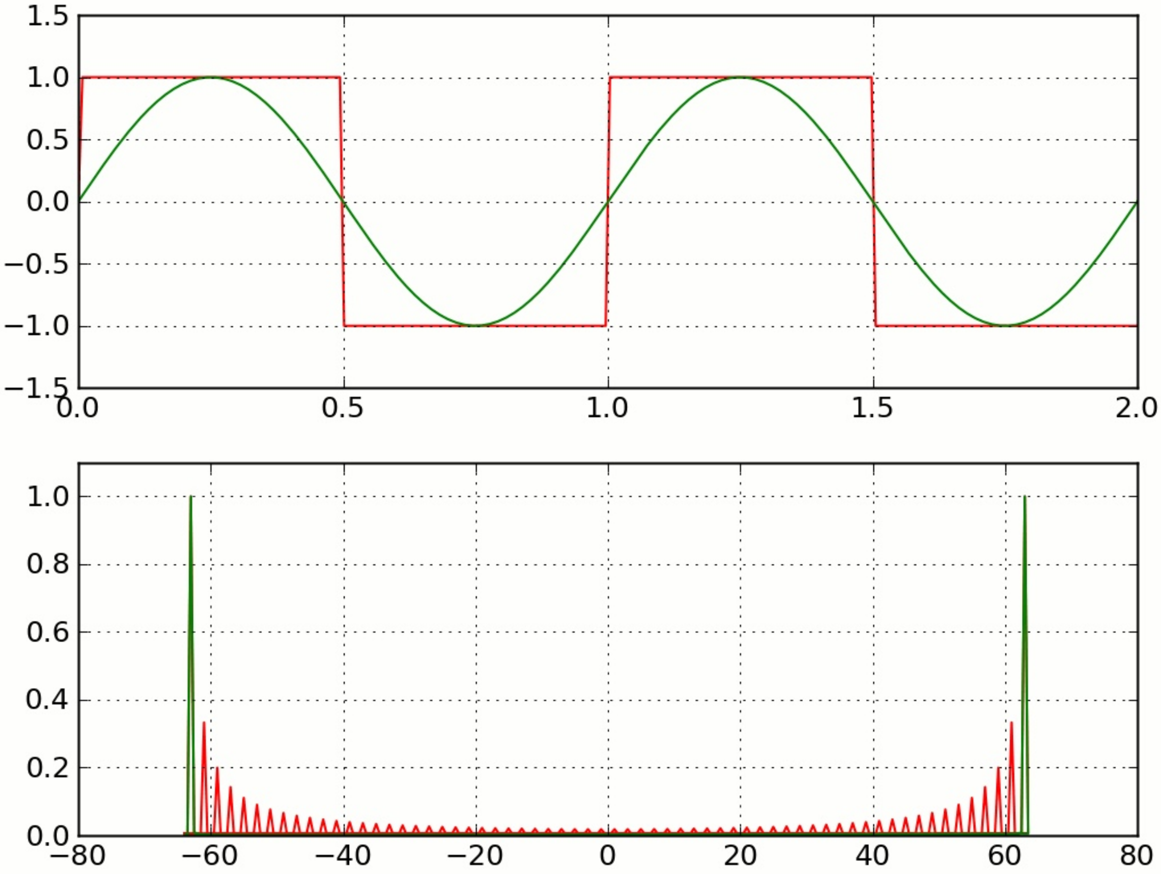
Question Does inner product (1) always exist?
Can we construct periodic signals f & g for which (1) does not exist?

Exercise: $$\int_0^1 \sin 2\pi x \, \sin 6\pi x \, dx = 0$$





1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0.0
0.5
1.0
1.5
2.0
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80



1.5
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2.0
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80

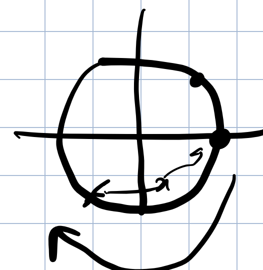
Periodic functions:

$f: \mathbb{R} \to \mathbb{R}$ $\qquad$ $f: \mathbb{C} \to \mathbb{C}$

$f: \mathbb{Z} \to \mathbb{Z}$

$f: \mathbb{R} \to \mathbb{R}$ is periodic if

$$f(t+T) = f(t) \quad \forall t \in \mathbb{R}.$$



Examples:

$f(x) = \sin x$ $\quad$ periodic $\quad 2\pi$.

$\sin 2\pi x$ $\quad$ period $\quad 1$.

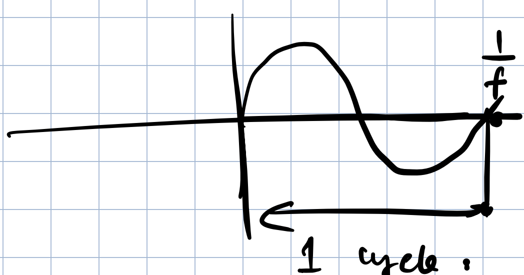
$\cos 2\pi x$ $\quad$ period $\quad 1$.

$\cos 2\pi f x$ $\quad$ period: $\quad \frac{1}{f}$.

Fundamental period is the smallest period.

frequency: no of cycles/revolutions per second (Hz)

$f(t) = 1 \quad \forall t$.



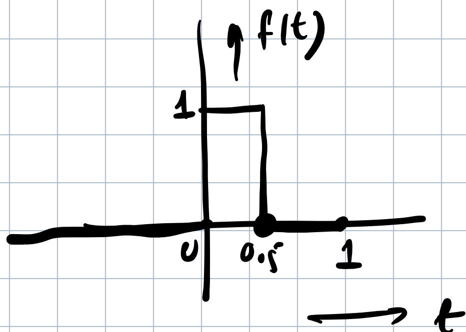
Dirichlet functions:

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{otherwise.} \end{cases}$$

Periodic with any rational period.

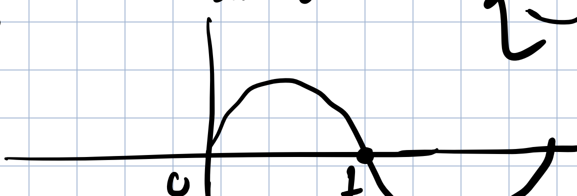
$$f(x+T) = f(x)$$

Continuous functions:



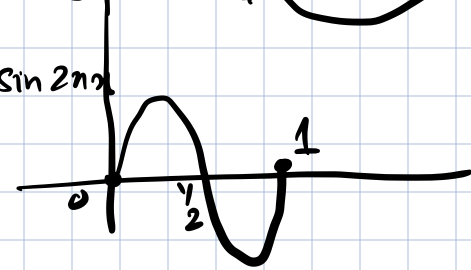
$\underbrace{\sin 2\pi x}_{\text{fundamental}} + \underbrace{\sin \pi x}$

$\sin \pi x$



$\left\{ \underbrace{\sin 2\pi x} + \underbrace{\sin 4\pi x} + \underbrace{\sin 6\pi x} \right.$

period: 2



$$\underbrace{\sin 2\pi x}_{} + \underbrace{\sin 4\pi x}_{}$$

fundamental period: 1

$$\underbrace{\sin 2\pi x}_{1\text{ sec}} + \underbrace{\sin 5\pi x}_{\frac{2}{5}\text{ sec.}}$$

$\sin 2\pi x \cdot \frac{5}{2}$

$\frac{2}{5}, \frac{4}{5}, \frac{6}{5}, \frac{8}{5}, \frac{10}{5}, \cdots$

$1, 2, 3, 4, 5, 6 \cdots$ fundamental period: 2.

$$\sin x + \underbrace{\sin \sqrt{2}\, x}_{} \quad (\text{not periodic}).$$

$\{2\pi, 4\pi, \cdots\}$

$\{\ ,\ \}$

$$\sin x + \sin \underbrace{1.414}_{} x$$

$\sin \sqrt{2} x \leftrightarrow \sin 1.414 x$

$$\underbrace{b_1 \sin 2\pi x}_{} + \underbrace{b_2 \sin 4\pi x}_{} + \underbrace{b_3 \sin 6\pi x}_{} + \underbrace{b_4 \sin 8\pi x}_{}$$

$$\underbrace{\cos 2\pi x}_{} + \underbrace{\cos 4\pi x}_{} + \cos 6\pi x + \cos 8\pi x + \cdots$$

$$+ \cos 2\pi m x$$

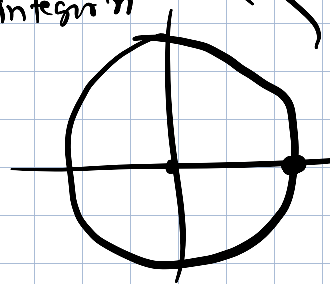
$$\frac{e^{2\pi i x} + e^{-2\pi i x}}{2} + \frac{e^{4\pi i x} + e^{-4\pi i x}}{2} + \cdots$$

$=$

$e^{2\pi i x}$ for

Complex plane.

$\boxed{e = -1}$ $\quad e = 1$ for any integer n



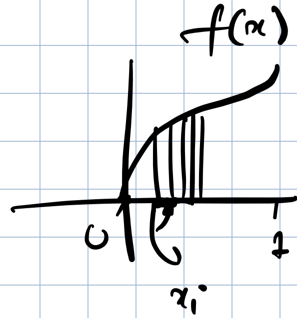
$f : \mathbb{R} \to \mathbb{C}$

$$f(x) = e^{2\pi i x} = \cos 2\pi x + i \sin 2\pi x.$$

complex exponential.

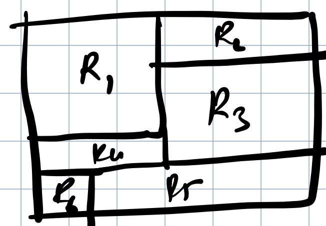
$$\boxed{\int_0^1 \sin 2\pi x \, dx} = 0 \qquad \int_0^1 \cos 2\pi x \, dx = 0$$

$$\int_0^1 e^{2\pi i x} \, dx = 0$$

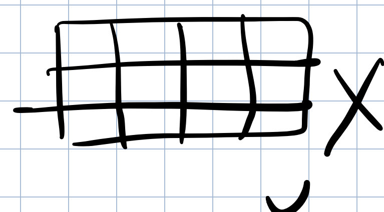


$$\int f(x)\,dx = \sum_i f(x_i)\,(x_{i+1} - x_i)$$

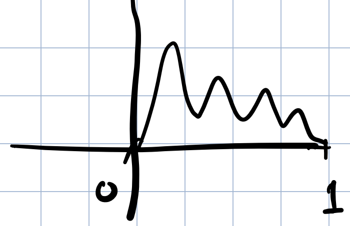
<u>Excercise</u>: Suppose R is a rectangle which is broken up into numerous smaller rectangles such that each of the smaller rectangles has at least one integer side.

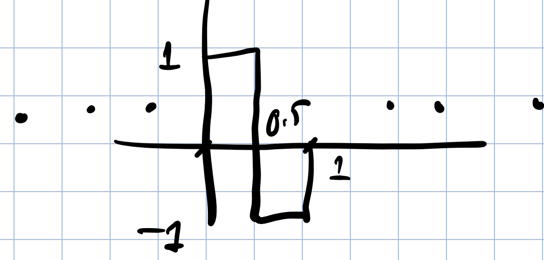


Show that the larger rectangle also has an integer side.



$$\underbrace{e^{2\pi i x} + e^{4\pi i x} + e^{6\pi i x}}$$





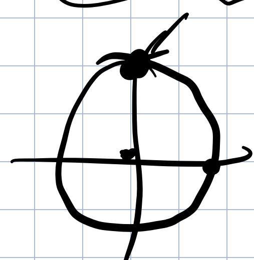
P = set of all periodic functions of period 1 $\quad f : \mathbb{R} \to \mathbb{C}$.

P forms a vector space over $\mathbb{C}$.

Inner product on P: $\quad x = \begin{bmatrix} i \\ 0 \end{bmatrix} \quad \dfrac{i\, e^{2\pi i x}}{}$

$$\underbrace{e^{-2\pi i x}} \qquad \int_0^1 f(x)\,\overline{g(x)}\,dx \qquad i^2 \qquad e^{2\pi i x + \pi i/2}$$

$$= \underbrace{e^{2\pi i \left(x + \frac{1}{4}\right)}}$$



$\left\{ \begin{array}{l} e^{2\pi i n x} \ \& \ e^{2\pi i m x} \ \text{ for } n \neq m \text{ integers} \\ \text{are orthogonal.} \end{array} \right.$

$$\int_0^1 e^{2\pi i n x}\, e^{-2\pi i m x}\, dx$$

$$= \int_0^1 \underbrace{e^{2\pi i (n-m) x}}\, dx = 0$$

since $n-m \neq 0$ is an integer.

Ex:

$$\left\{ \begin{array}{ll} \displaystyle\int_0^1 \sin 2\pi m x \, \sin 2\pi n x \, dx = 0 & m \neq n. \\ \displaystyle\int_0^1 \sin 2\pi m x \, \cos 2\pi n x \, dx = 0 & \\ \displaystyle\int_0^1 \cos 2\pi n x \, \cos 2\pi m x \, dx = 0 & m \neq n. \end{array} \right.$$

① If $c_1 \sin 2\pi x + c_2 \sin 4\pi x$

$= a_1 \cos 10\pi x$

$+ a_2 \cos 12\pi x$

$\forall x.$

$\exists\ \sin 2\pi x + (-1)\sin 2\pi x = 0$

$$\begin{cases} 1\,\underbrace{(1+x^2)}_{\to f_1} + 1\,\underbrace{(1+x)}_{\to f_2} \\ +1\,\underbrace{(x+x^2)}_{\to f_3} \\ \quad -2\underbrace{(1+x+x^2)}_{\to f_4} = \\ c_1 f_1(x) + c_2 f_2(x) + c_3 f_3(x) + c_4 f_4(x) = 0 \end{cases}$$

①

$$c_1 \underbrace{\int_0^1 \sin 2\pi x \sin 2\pi x}_{} + \underbrace{c_2 \int_0^1 \sin 4\pi x \sin 2\pi x}_{0} = a_1 \underbrace{\int_0^1 \cos 10\pi x \sin 2\pi x}_{0} + a_2 \underbrace{\int_0^1 \cos 12\pi x \sin 2\pi x}_{0}$$

$$c_1 = 0$$

$$\begin{cases} \alpha_1 v_1 + \alpha_2 v_2 = 0 \\ \alpha_1 \underbrace{v_2^T v_1}_{} + \alpha_2 v_2^T v_2 = 0 \end{cases}$$

Any periodic function of period 1 can be expressed as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i n x}$$

$n \in \mathbb{Z}$.

$\{e^{2\pi i n x}\}_{n=-\infty}^{\infty}$ is an orthogonal basis of P.

$\underline{v}_1 \quad \underline{v}_2 \quad \underline{v}_3$

$$\underline{v} = \alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \alpha_3 \underline{v}_3$$

$$\begin{bmatrix} \underline{v}_1 & \underline{v}_2 & \underline{v}_3 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \underline{v}$$

$$\underline{v}_1^T \underline{v} = \underbrace{\alpha_1 \underline{v}_1^T \underline{v}_1} + \cancel{\alpha_2 \underline{v}_1^T \underline{v}_2} + \cancel{\alpha_3 \underline{v}_1^T \underline{v}_3}$$

If $\underline{v}_1, \underline{v}_2, \underline{v}_3$ orthogonal, then

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} \underline{v}_1^T \\ \underline{v}_2^T \\ \underline{v}_3^T \end{bmatrix} \underline{v} = \begin{bmatrix} v_1^T v \\ v_2^T v \\ v_3^T v \end{bmatrix}$$

By analogy,

$$C_n = (f, e^{2\pi i n x}) = \int_0^1 f(x)\, e^{-2\pi i n x}\, dx$$

<u>Solutions to differential equations</u>:

$$\frac{dy}{dx} = -y \qquad \left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} + y = 10_x$$

Order of a diff eqn is the highest derivative appearing in the differential eqn.

Linear constant coeff homogenous differential equations

Linear: only linear combinations of

$$\frac{d^n y}{dx^n}, \frac{d^{n-1} y}{dx^{n-1}}, \ldots, \frac{dy}{dx}, y \quad \text{appear}$$

in the differential equation.

$$\boxed{}\frac{d^2y}{dx^2} + \boxed{10}\frac{dy}{dx} - \boxed{}y = 0$$

coeffecients

Property. the set of all solutions of linear homogenous differential equation form a vector space.

$$\frac{d^n(\alpha y)}{dx^n} = \alpha \frac{d^n y}{dx^n}$$

$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$$

$$x\frac{d^2(\alpha y)}{dx^2} + \frac{d(\alpha y)}{dx} - \alpha y = \alpha\underbrace{\left[x\frac{d^2y}{dx^2} + \frac{dy}{dx} - y\right]}$$

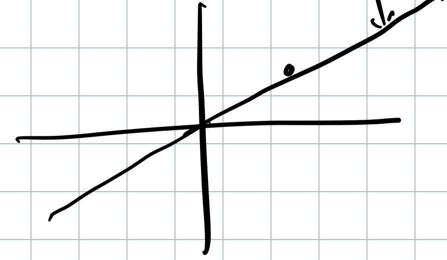
$$= 0$$

If y_1 & y_2 are solutions, so is $y_1 + y_2$

(Excercise)

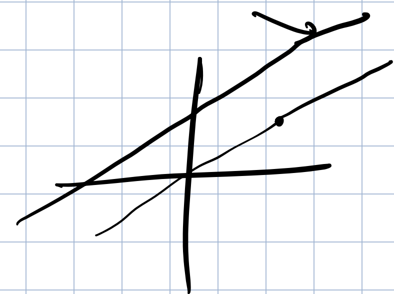
$y_1 - y_0$

$$\frac{dy}{dx} = y$$



sol^n: $y = ke^x$.

Set of all sol^{n}s: all scalar multiples of e^x.



$$\frac{d^2y}{dx^2} = -y$$

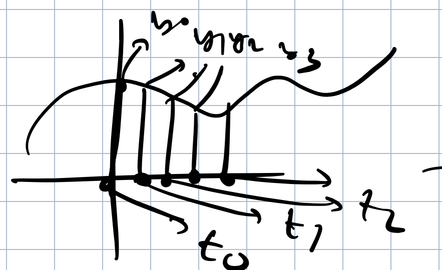
$$y = A\sin x + B\cos x$$

$$\frac{dy}{dx} = y+2 \qquad y = ke^x - 2$$

Exercise: Consider the set of all sequences F_n that satisfy

$$F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 2.$$

derivative $\longleftrightarrow$ finite difference



$$y(x) \longleftrightarrow y_n$$

$$\frac{dy}{dx} \longleftrightarrow \frac{y_{n+1}-y_n}{t_{n+1}-t_n}$$

differential eq$^{\underline{n}}$ $\longleftrightarrow$ recurrence relation.

__Explanation:__ F_n is completely determined by F_0 & F_1

In general, $F_n = c_n F_0 + d_n F_1$

$F_2 = F_0 + F_1$, $F_3 = F_2 + F_1 = F_0 + F_1 + F_1 = F_0 + 2F_1$

$F_4 = F_3 + F_2 = F_0 + 2F_1 + F_0 + F_1 = 2F_0 + 3F_1$

__Case 1:__ $F_0 = 1$, $F_1 = 0$

Let the obtained solution be $\begin{bmatrix}1\\0\end{bmatrix}$

$\tilde{F}_n$

__Case 2:__ $F_0 = 0$, $F_1 = 1$

Let the obtained solution be $\begin{bmatrix}0\\1\end{bmatrix}$

$\hat{F}_n$

An arbitrary solution is $\alpha \tilde{F}_n + \beta \hat{F}_n$.

__Exercise:__ Is $\mathbb{R}$ a vector space over $\mathbb{Q}$?

__Answer:__ Yes.

are $\sqrt{2}$ and 1 linearly independent?

$$\underbrace{\alpha\sqrt{2} + \beta.1 = 0}_{\text{not possible unless } \alpha, \beta = 0.} \qquad \alpha, \beta \in \mathbb{Q}.$$